



COMP 2067

Introduction to

Formal Reasoning

Lecture 1 Introduction



Self-introduction

- **Dr. Kweh Yeah Lun**
- Office: **DB33** (by appointment)
 - Alternatively, you can also get in touch with me via **Teams** or **email**
- email: kweh.yeahlun@nottingham.edu.my
- Bachelor degree: Computer Science
- Master degree: Computer Science
- Ph.D. in Parallel and Distributed Computing (graduated 2011 from UPM)
- Research Interest: Distributed algorithm, parallel algorithm, distributed and parallel computing, quantum computing, etc.



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Synopsis

- This module is an **introduction to formal reasoning** with applications in **program verification** and **mathematics**.
- We will use an interactive proof system like **Lean 3** to learn how to make precise statements and how to verify them using formal proofs.
- At the same time, we develop skills in **informal reasoning**.
- Topic covered are: **propositional logic, classical principles, predicate logic, the theories of Booleans, natural numbers, lists and trees and verification of simple algorithms.**



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Topics

- Propositional logic
- Classical logic
- Predicate logic
- The Booleans
- The natural numbers
- Lists and Trees
- Lean 3 will be used



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Assessment

- **Coursework: 25%**
 - Lab tests
 - Quiz
- **Final Exam: 75%**
 - 2 hours



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Mode of Delivery

- Lecture
- Tutorial -> Lecture
- Lab



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Communication Channels

- Lecture, lab or tutorial
- Moodle
- Email
- Teams Chat
- Teams Meeting



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Main Reference

- https://people.cs.nott.ac.uk/psztxa/comp2065.23-24.ifr-notes/_build/html/introduction.html



Additional References

- Theorem Proving in Lean
 - [Theorem Proving in Lean — Theorem Proving in Lean 3.2 3.0 documentation \(leanprover.github.io\)](https://leanprover.github.io/theorem_proving_in_lean3/)
- The Hitchhiker's Guide to Logical Verification



- We are using the Lean prover, available on windows, linux and OSX.
- Make sure to install “Lean3” not “Lean4” - The lectures are in Lean3.
- Lean can be used online via a browser (supporting javascript).
- Lean can be used via emacs (with lean-mode) or Visual Studio Code (recommended).
- Online version of Lean 3
 - <https://leanprover-community.github.io/lean-web-editor/#code=--%20start%20your%20code%20below!%0A>